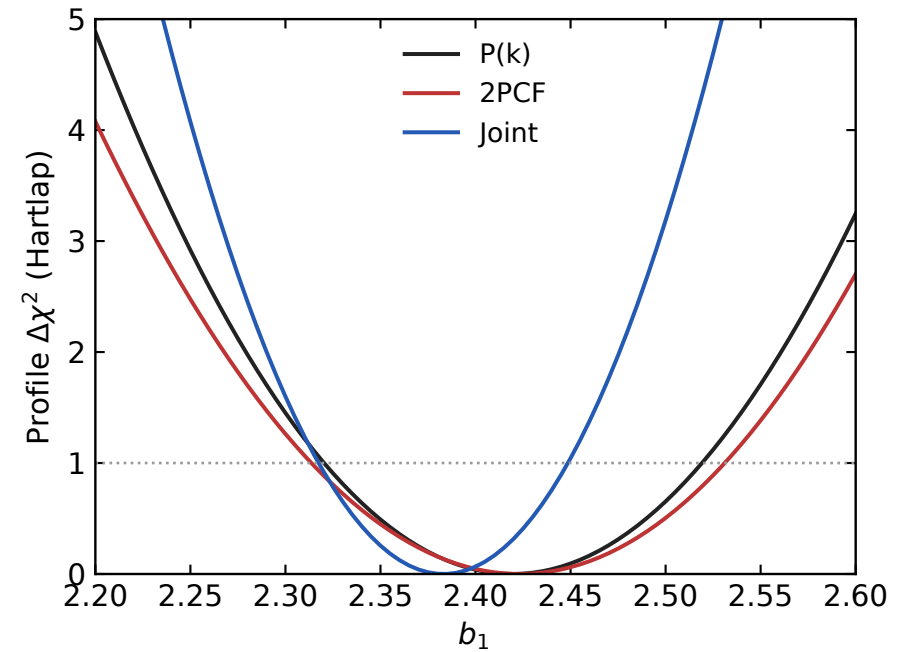


Why does the joint fit prefer a lower b_1 ?

Frozen 9.18 data selection; mandatory hybrid GIC; all main estimates below are maximum likelihood.

Fit	b_1	f_{NL}	Hartlap χ^2
P(k)	2.419496	-11.966	1.111
2PCF	2.423165	-2.065	3.073
Joint	2.383164	-1.354	4.180



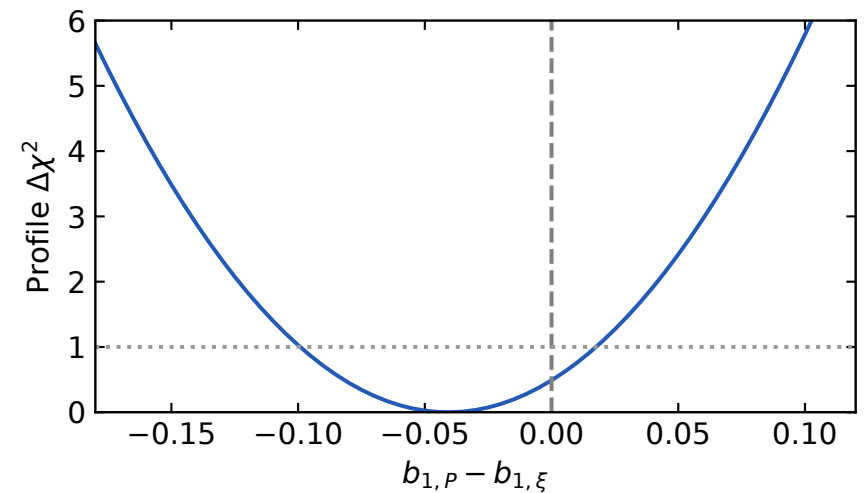
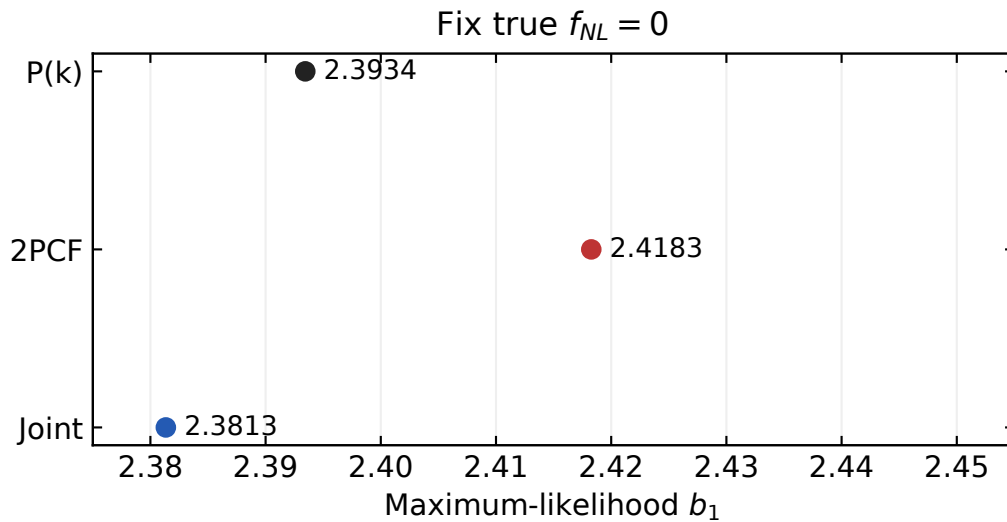
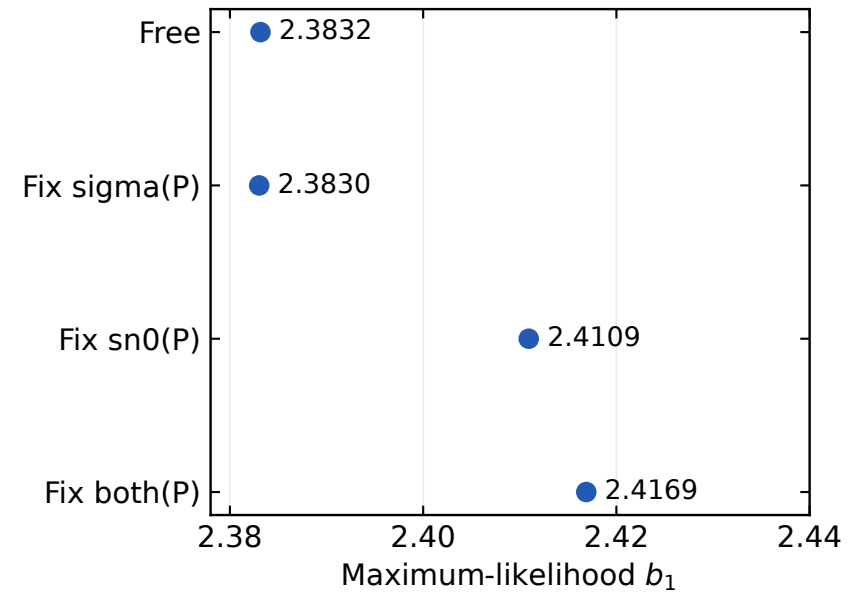
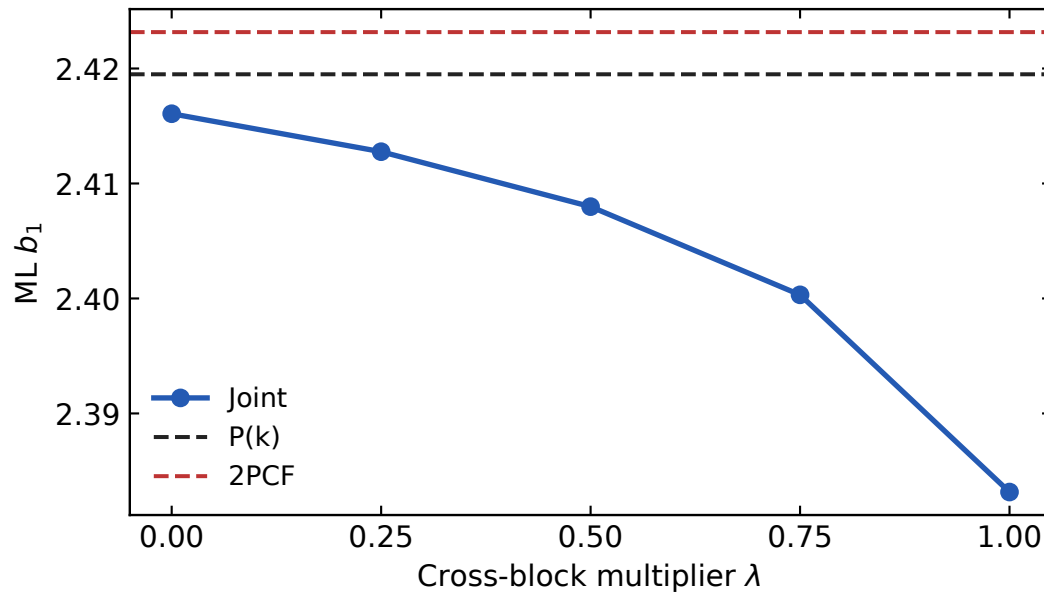
Primary finding

Strong P- ξ correlations turn a coherent ξ_0 residual into a lower shared b_1 .
The P(k) constant term sn_0 absorbs part of the corresponding P0 amplitude change.

Contract: 13 P0 + 9 P2 + 26 ξ_0 + 26 ξ_2 bins; $k_{\max}=0.08$; $s_{\min}=50$; BAO mask 80-120.
Data: mean of 25 Abacus phases. Formal fit covariance: C_{single} , including the full cross block.
No new GSM parameter, no changed formal contour, and no criterion requiring joint b_1 to lie between fits.

Cross covariance and parameter coupling

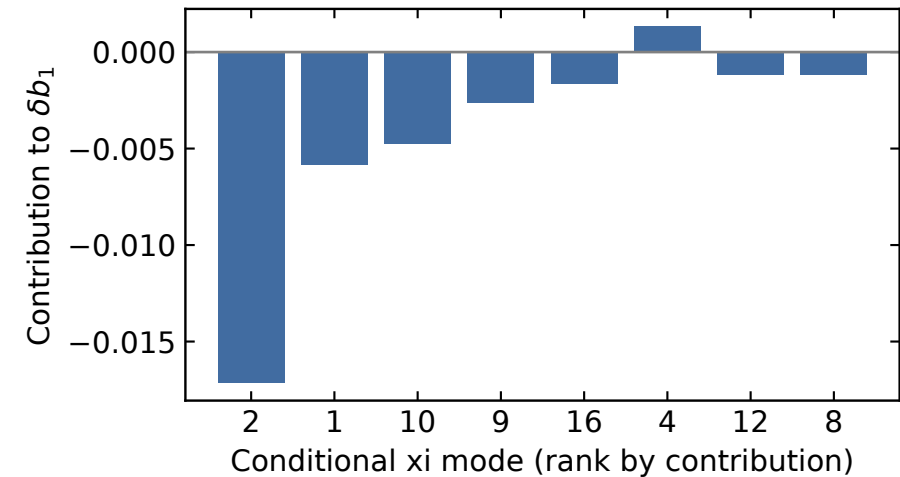
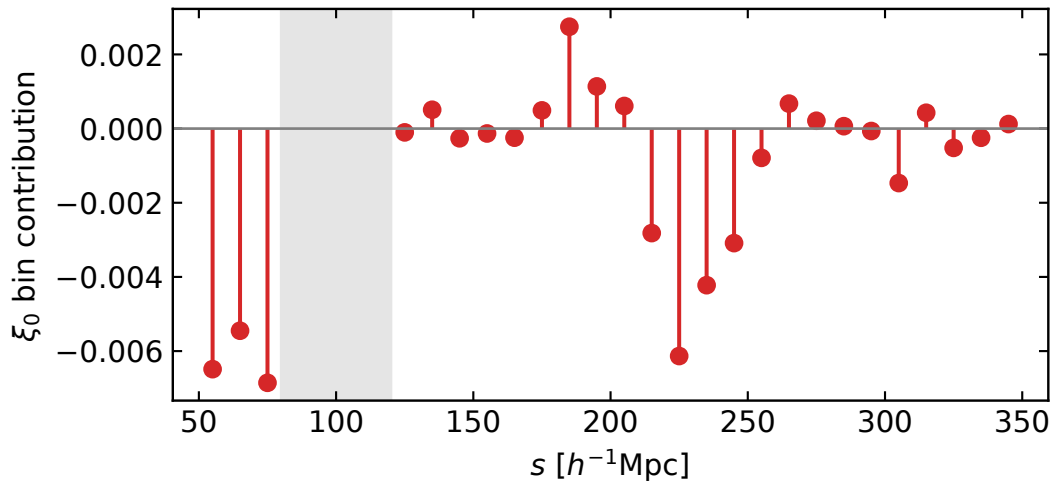
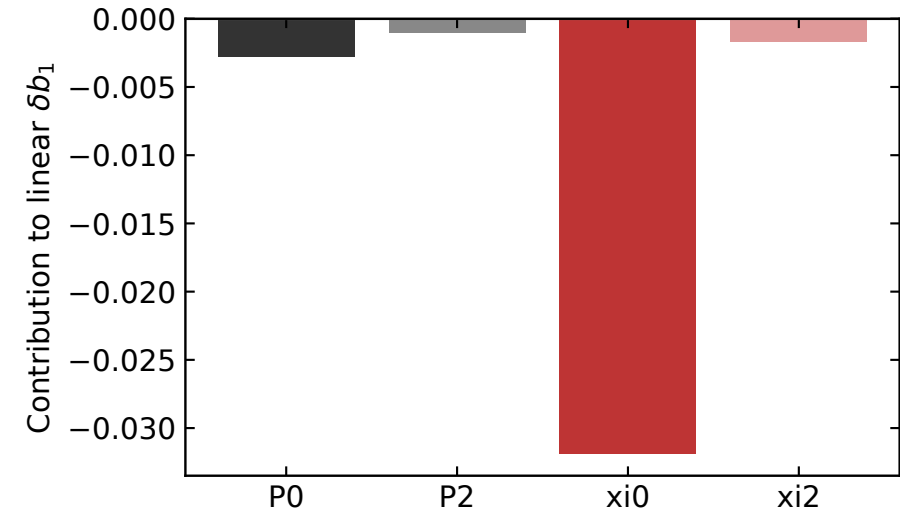
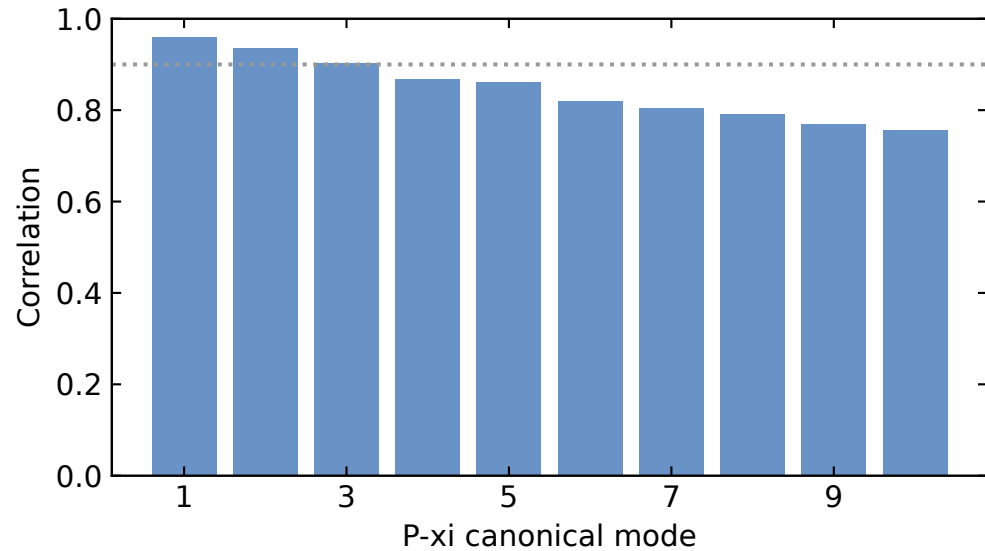
Both covariance marginals remain fixed as the cross block is multiplied by lambda.



Cross removal and fixed nuisance values are diagnostics, not proposed fixes. Covariance interpolation remains positive definite.
 Splitting b_1 improves raw C_{single} χ^2 by only 0.528; C_{single} error bars must not be confused with the precision of a 25-phase mean.
 Task 4.3.2 | hybrid + GIC | EZmock1000 | diagnostic results

Conditional residuals explain the displacement

Linearization is evaluated at the P(k)-only optimum; GIC is included in every derivative.



Linear prediction $b_1 = 2.382122$; optimized joint $b_1 = 2.383164$. Schur identity error $< 1e-8$.

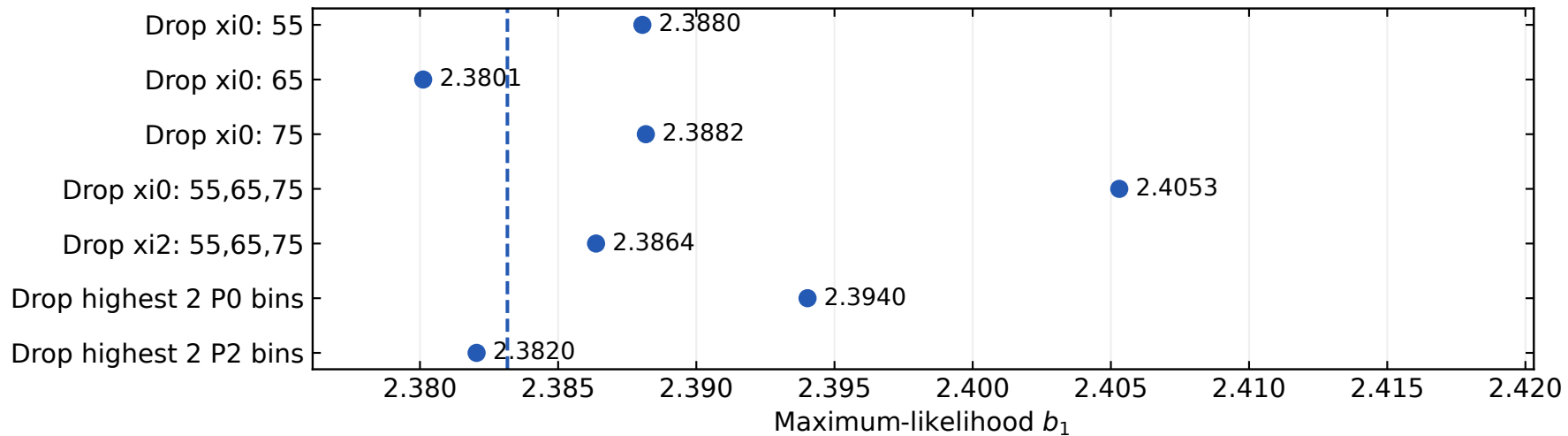
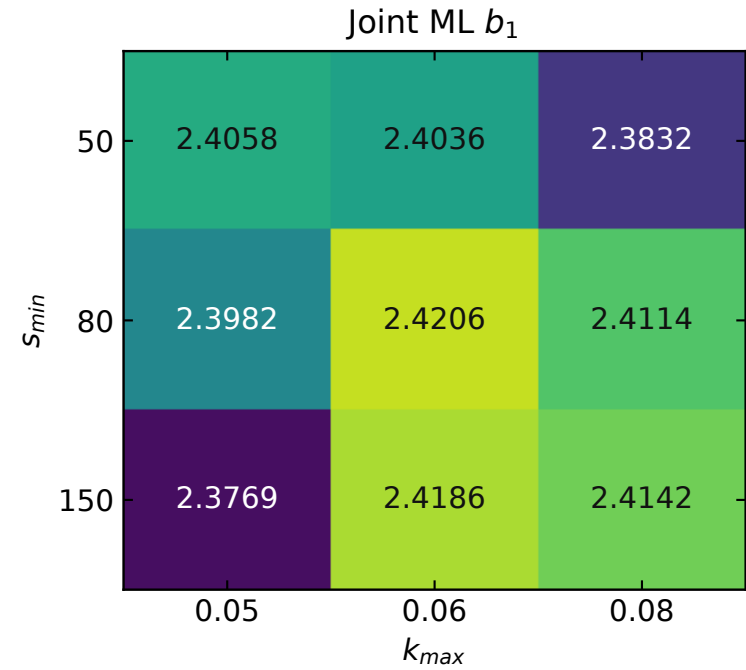
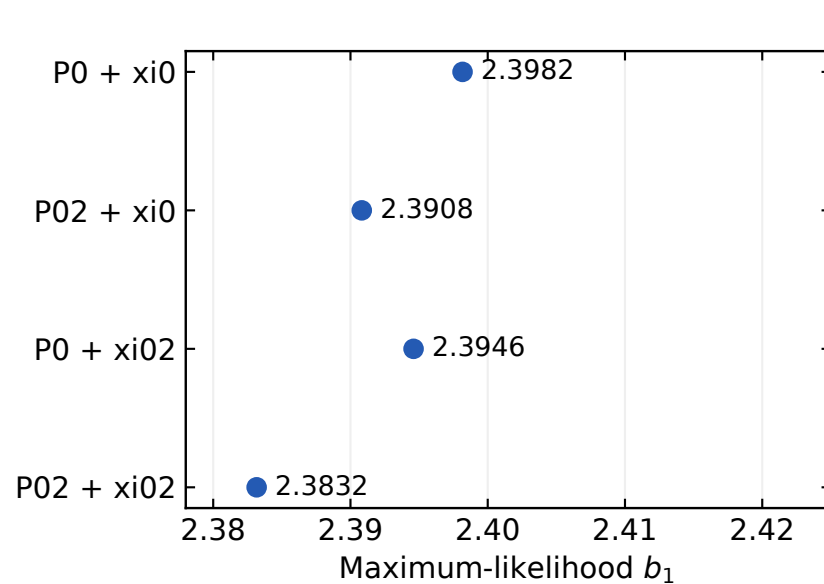
Attribution is a local, correlated decomposition, not a set of independent causal effects. xi0 dominates this decomposition.

The conditional modes are combinations of bins; the strongest canonical correlation is 0.9598.

Task 4.3.2 | hybrid + GIC | EZmock1000 | diagnostic results

Which multipoles and scales matter?

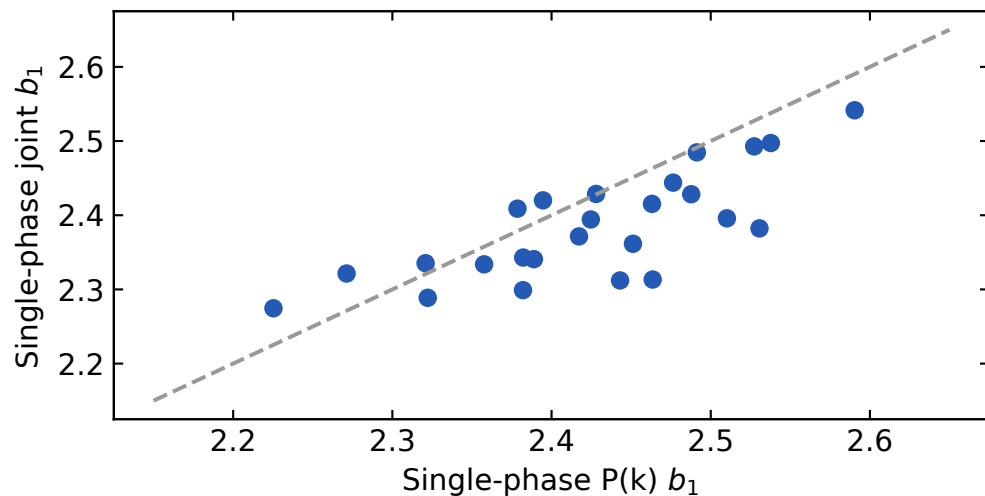
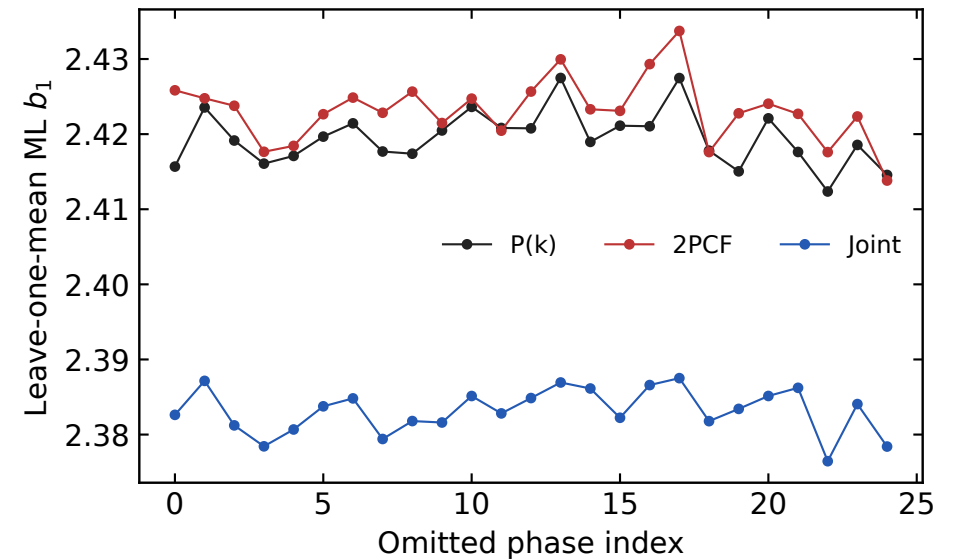
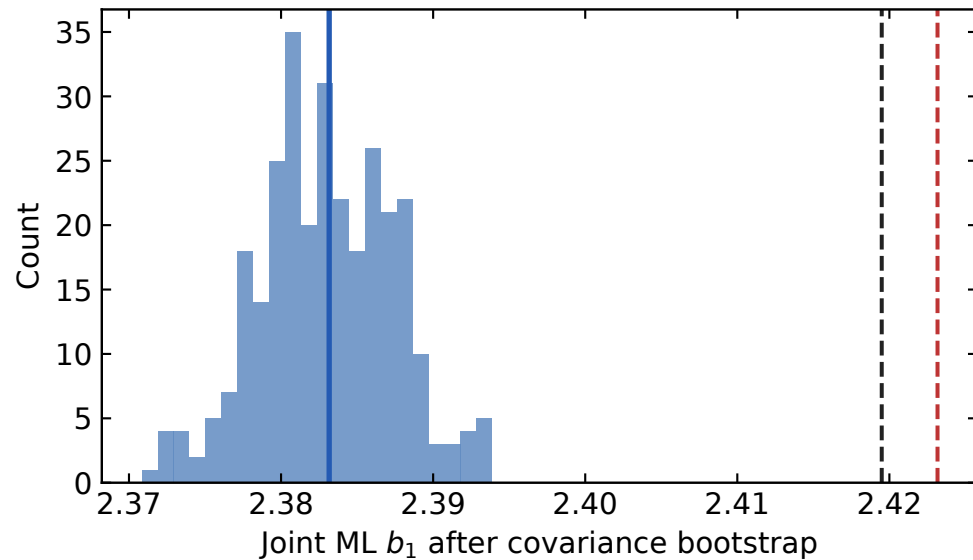
Each retained data vector uses the matching covariance submatrix before inversion.



The dashed line is the full-contract joint fit. Changes in fitted b_1 after cuts do not by themselves validate those cuts. Because $80 \leq s < 120$ is masked, $s_{min}=80$ and $s_{min}=100$ retain exactly the same bins. k and s cuts are not equivalent Fourier cuts.
Task 4.3.2 | hybrid + GIC | EZmock1000 | diagnostic results

The shift survives covariance and phase resampling

300 paired-mock covariance bootstraps; 25 phase fits and 25 leave-one-phase-out means.



Stability summary

Bootstrap sigma(b_{1_joint}) = 0.0044

Joint below both fits: 300/300 bootstraps

Joint below both fits: 25/25 leave-one means

Single-phase residual trace / 74 = 1.049

Projected b_1 variance / covariance prediction = 1.318

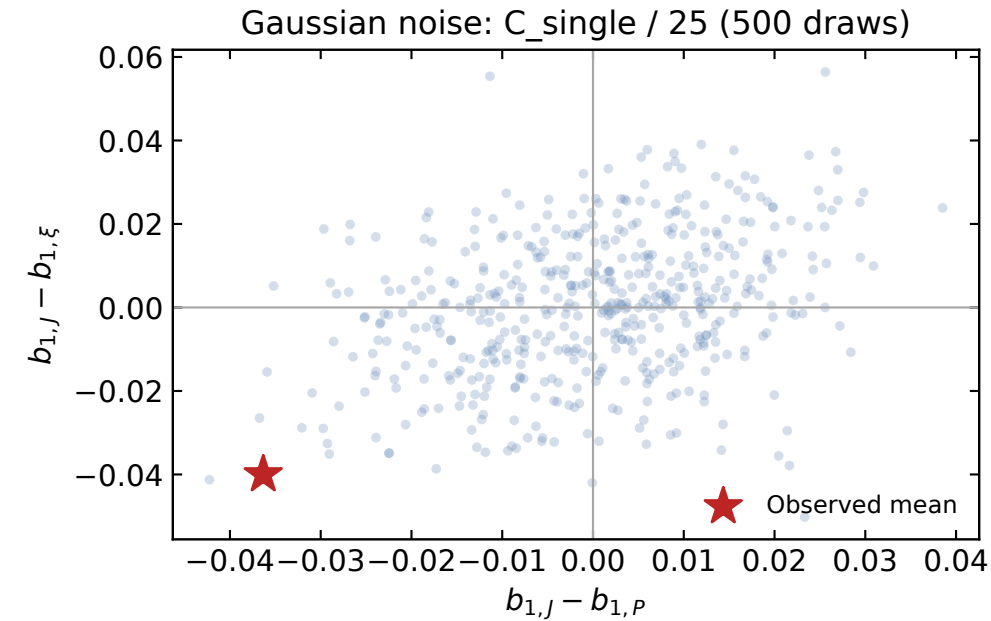
Bootstrap samples estimate sensitivity to the finite 1000-mock covariance, conditional on this mock ensemble.

Leave-one means are strongly correlated. Phase fits use the same mean forward operator; the 25-phase covariance is never inverted.

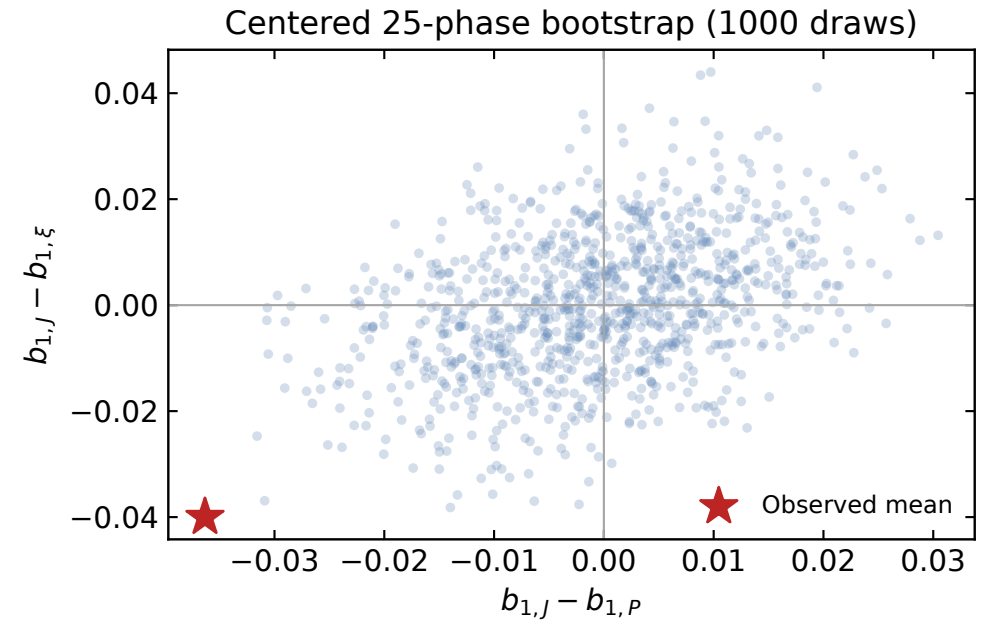
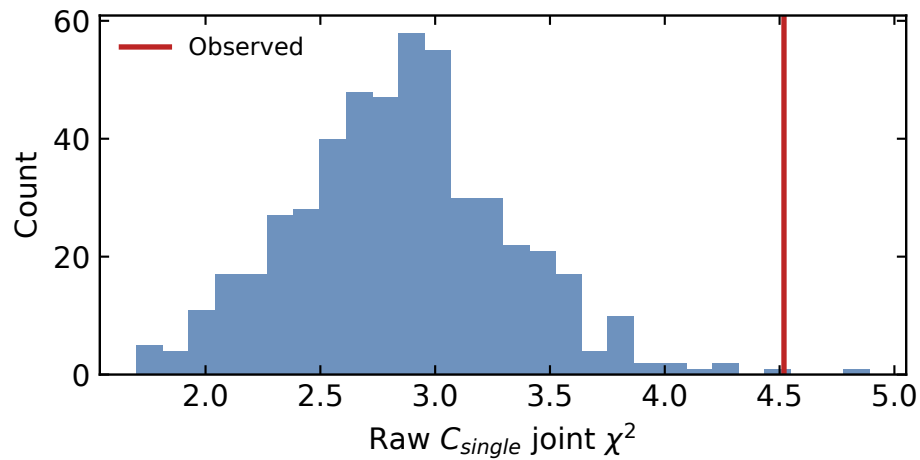
Task 4.3.2 | hybrid + GIC | EZmock1000 | diagnostic results

Is the observed displacement a usual noise effect?

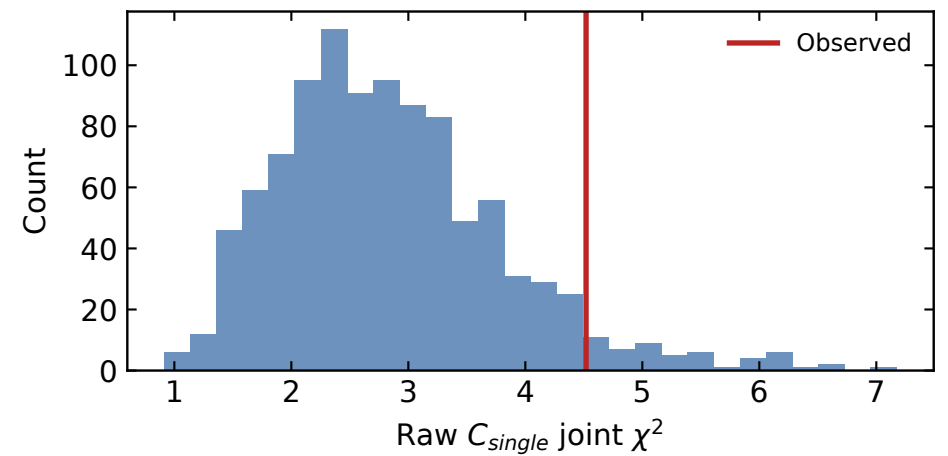
Fit covariance remains C_{single} ; the noise used to generate a 25-phase mean is explicitly separate.



2D tail: 6/500 draws
Joint below both fits: 28.2%



2D tail: 0/1000 draws (resolution limited)
Joint below both fits: 32.4%



Tail statistic uses both correlated ML differences. Empirical frequencies include +1 correction and finite Monte Carlo resolution.
Gaussian mean noise assumes independent phases and the chosen covariance. Phase bootstrap has only 25 input phases; neither test proves model correctness.
Task 4.3.2 | hybrid + GIC | EZmock1000 | diagnostic results

Observation-operator checks and remaining gaps

Shell averaging is quantified; missing phase windows and differing conventions remain explicit.

Fit	Bin center	Shell average	Delta b1
P(k)	2.419496	2.419496	-0.000000
2PCF	2.423165	2.431787	+0.008622
Joint	2.383164	2.387379	+0.004215

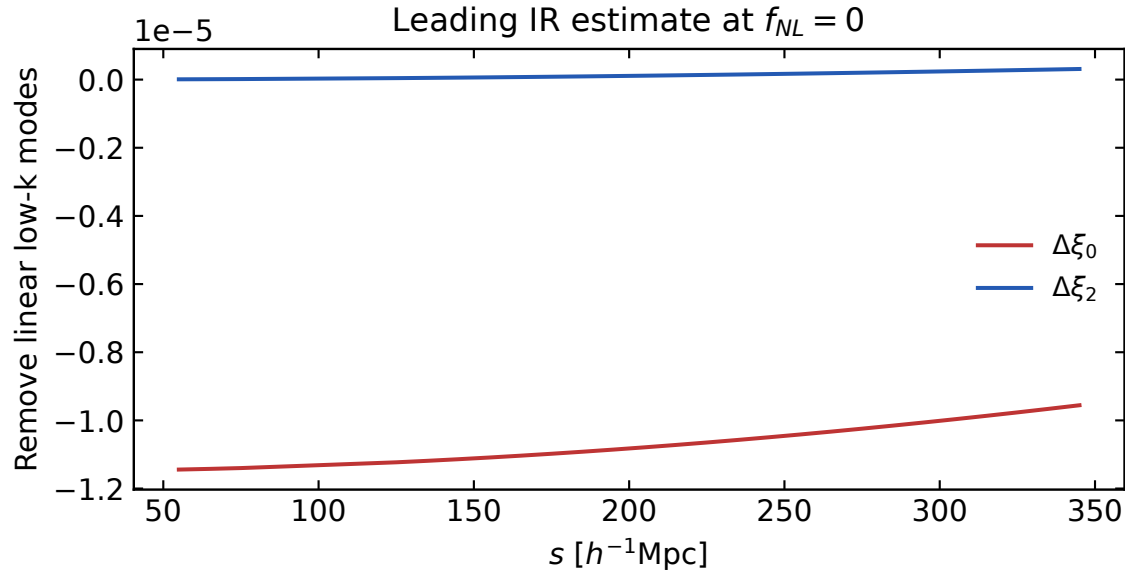
Verified input agreement

25/25 phases: same FKP summary and zeff

FKP P0 = 10000 on both sides

P window: only ph000 exists

P LOS: local; xi LOS: midpoint



Low-k support audit

P theory cutoff: $k_{\text{box}} = 0.003142$

Hybrid FFT starts at $k = 1\text{e-}05$

Linear-only IR response: Delta b1 =

-0.00558 (matched scalar GIC included)

This is not a full nonlinear finite-box calculation.

What the diagnostics establish

EZmock1000 is the sole covariance used for the conclusions and follow-up recommendations.

Established mechanism

Cross correlations amplify a coherent ξ_0 residual.
A local linear response reproduces the joint b_1 shift.

Important coupling

Fixing sn_0 at the P-only optimum moves joint b_1 upward.
Fixing $f_{NL}=0$ or $\sigma_{s,P}$ alone does not remove the shift.

Robustness checks

The effect survives 300 covariance bootstraps and all 25 leave-one means. Shell averaging has a small effect.

Interpretation

Outside both marginal optima is allowed for correlated fits.
For this 25-phase mean, closure tests still flag a residual worth investigating; formal C_{single} intervals hide this precision.

Next targeted work

Audit matched P_0/ξ_0 broadband and shot-noise response, using the identified scale ranges and conditional modes.
Complete the P mean-window and finite-box operator checks.

Avoid premature fixes

Do not tune the covariance, fix sn_0 by preference, or choose scale cuts merely to put joint b_1 between the two fits.
Keep the formal hybrid+GIC baseline frozen.